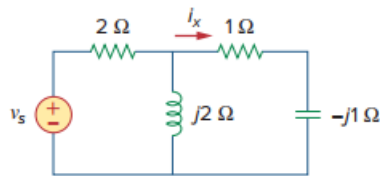


Problem

Find $v_s(t)$ in the circuit of Fig. 9.56 if the current i_x through the $1\text{-}\Omega$ resistor is $8 \sin 200t$ A.

Figure 9.56:



Step-by-step solution

Step 1 of 5

The current flowing through the $1\text{ }\Omega$ resistor is,

$$i_x = 8 \sin 200t \text{ A}$$

It is known that,

$$\sin \omega t = \cos(\omega t - 90^\circ)$$

Therefore, the current i_x is,

$$i_x = 8 \cos(200t - 90^\circ) \text{ A}$$

The phasor domain representation of current i_x is,

$$I_x = 8 \angle -90^\circ \text{ A}$$

Step 2 of 5

Refer to Figure 9.56 in the textbook.

Draw the circuit diagram with current directions.

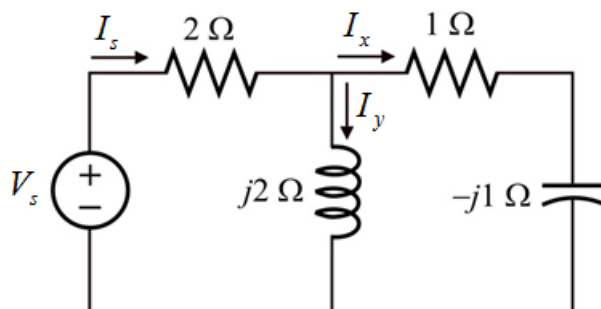


Figure 1

Step 3 of 5

Write the current expression I_x using current division rule.

$$I_x = \left(\frac{j2}{j2+1-j1} \right) I_s$$

Substitute $8\angle -90^\circ$ A for I_s in the equation.

$$8\angle -90^\circ = \left(\frac{j2}{j2+1-j1} \right) I_x$$

$$8\angle -90^\circ = (1+j) I_x$$

$$I_x = \frac{8\angle -90^\circ}{1+j}$$

$$I_x = (-4-j4) \text{ A}$$

Step 4 of 5

Determine the current flowing through the $j2 \Omega$ inductor.

$$I_y = I_s - I_x$$

Substitute $8\angle -90^\circ$ A for I_s and $(-4-j4)$ A for I_x in the equation.

$$\begin{aligned} I_y &= (-4-j4) - (8\angle -90^\circ) \\ &= (-4-j4) - (-j8) \\ &= (-4+j4) \text{ A} \end{aligned}$$

Step 5 of 5

Apply Kirchhoff's Voltage Law to left side loop.

$$-V_s + (2)I_s + (j2)I_y = 0$$

Substitute $(-4-j4)$ A for I_s and $(-4+j4)$ A for I_y in the equation.

$$\begin{aligned} -V_s + (2)(-4-j4) + (j2)(-4+j4) &= 0 \\ -V_s + (-8-j8) + (-8-j8) &= 0 \\ -V_s - 16 - j16 &= 0 \\ V_s &= -16 - j16 \\ &= 22.63\angle -135^\circ \text{ V} \end{aligned}$$

The time domain expression for source voltage is,

$$v_s(t) = 22.63 \cos(200t - 135^\circ) \text{ V}$$

It is known that,

$$\cos \omega t = \sin(\omega t + 90^\circ)$$

The expression for source voltage is,

$$\begin{aligned} v_s(t) &= 22.63 \sin(200t - 135^\circ + 90^\circ) \text{ V} \\ &= 22.63 \sin(200t - 45^\circ) \text{ V} \end{aligned}$$

Therefore, the value of source voltage, $v_s(t)$ is $\boxed{22.63 \sin(200t - 45^\circ) \text{ V}}$.